Severity & Case Classification

Classifying adverse drug reactions by severity, seriousness and expectedness, and judging their clinical impact across multiple cases

PHARMACOVIGILANCE INTERNSHIP | TASK 4

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Goal of this task: classify each ADR case by **severity**, **seriousness** and **expectedness**, decide how urgent it is, and rank several cases by clinical impact. This guide gives the definitions, a decision matrix, a workflow, and six worked cases.

1. Four Ideas That Must Stay Separate

Dimension	Question it answers	Judged against	Typical answer
Severity	How intense is the event?	A clinical scale (Hartwig, CTCAE)	Mild, moderate, severe
Seriousness	Does the outcome meet a regulatory criterion?	ICH E2A criteria	Serious or non-serious
Expectedness	Is it already in the product information?	Reference safety information	Expected or unexpected
Causality	Did the drug cause it?	WHO-UMC, Naranjo (Task 3)	Certain to Unlikely

Key point

Each dimension is decided on its own. A **mild** event can be **serious**, a **severe** event can be **non-serious**, and an **expected** event can still be **serious**.

2. Severity: How Intense Is It?

Severity describes intensity and clinical burden. The **Hartwig-Siegel** scale has seven levels that group into mild, moderate and severe.

Level	Description	Group
1	ADR occurred but needed no change in treatment.	Mild
2	Suspected drug held, stopped or changed. No antidote or other treatment, no longer hospital stay.	Mild
3	Drug changed and/or an antidote or other treatment needed. No longer hospital stay.	Moderate
4	(a) Level 3 plus hospital stay extended by at least a day, or (b) the ADR was the reason for admission.	Moderate
5	Level 4 plus intensive medical care needed.	Severe
6	ADR caused permanent harm to the patient.	Severe
7	ADR directly or indirectly led to death.	Severe

*In oncology and trials, **CTCAE** grades are common: Grade 1 mild, 2 moderate, 3 severe or medically significant, 4 life-threatening, 5 death. Severity scales differ, so use the one your organisation specifies and name it in the report.*

3. Seriousness: Does the Outcome Meet a Criterion?

An event is serious if it meets **any one** of the ICH E2A criteria. Memory hook from Task 1: **D-L-H-D-C-M**.

Criterion	Meaning	Note
D Death	The event resulted in death, with a suspected drug link.	Death is an outcome; record the cause too.
L Life-threatening	The patient was at **immediate** risk of death at the time of the event.	Not an event that *might* have been fatal if worse.
H Hospitalisation	Inpatient admission, or a longer stay, because of the event.	Planned admission for a pre-existing condition, and an ER visit without admission, do not count.
D Disability	Persistent or significant disruption of normal life functions.	Not minor, short-lived inconvenience.
C Congenital anomaly	Birth defect after exposure, including exposure through a parent.	Include pregnancy exposures.
M Medically important	May jeopardise the patient or need intervention to prevent the outcomes above.	Examples: allergic bronchospasm treated in the ER, convulsions without admission, blood disorders.

Severity vs seriousness in practice

Event	Severity	Serious?	Why
Severe headache, treated at home, gone in a day	Severe	No	No seriousness criterion met.
Moderate rash needing admission for IV treatment	Moderate	**Yes**	Hospitalisation.
Anaphylaxis treated in the ER with adrenaline, no admission	Severe	**Yes**	Life-threatening or medically important.
Seizure after a drug, over in minutes, no admission	Moderate	**Yes**	Medically important event.
Planned knee surgery admission, unrelated to the drug	-	No	Planned admission, not caused by an AE.
Mild nausea for two days	Mild	No	No criterion met.

4. Expectedness: Is It Already Known?

An adverse reaction is **unexpected** if its nature or severity is not consistent with the product information. Judge it against the **reference safety information**, not against personal experience.

Setting	Reference document
Marketed medicine	Approved label: SmPC, package insert, or company core data sheet (the version in force).
Clinical trial	Investigator's Brochure (or the approved label if the product is already authorised).

Label lists	Case reports	Expectedness	Reason
Hepatitis	Fulminant hepatic failure	**Unexpected**	More severe or more specific than listed.
Acute renal failure	Interstitial nephritis	**Unexpected**	A more specific, new finding.
Dizziness	Dizziness	Expected	Listed.
Nothing about the event; only another drug in the same class has it	The event	**Unexpected**	A class effect counts only if the label says so for this drug.
Not mentioned anywhere	A new event	**Unexpected**	Not listed.

5. Classification Matrix and Reporting Pathway

	EXPECTED (listed)	UNEXPECTED (not listed, or worse)
SERIOUS	**Serious + Expected** Expedited in the EU (15 days); periodic report in the US. Track how often it occurs. **Medium to high impact.**	**Serious + Unexpected** **Highest priority.** Expedited alert report, urgent medical review, signal evaluation, possible label change.
NON-SERIOUS	**Non-serious + Expected** Routine. Aggregate and periodic reports; trend monitoring. **Low impact.**	**Non-serious + Unexpected** Periodic report (EU: 90 days). Keep for signal detection; escalate if cases accumulate. **Low to medium impact.**

SUSAR

Suspected + **U**nexpected + **S**erious **A**dverse **R**eaction (used in clinical trials). 'Suspected' means there is a reasonable possibility that the drug caused the event. Causality still matters: an unrelated serious event is not a SUSAR.

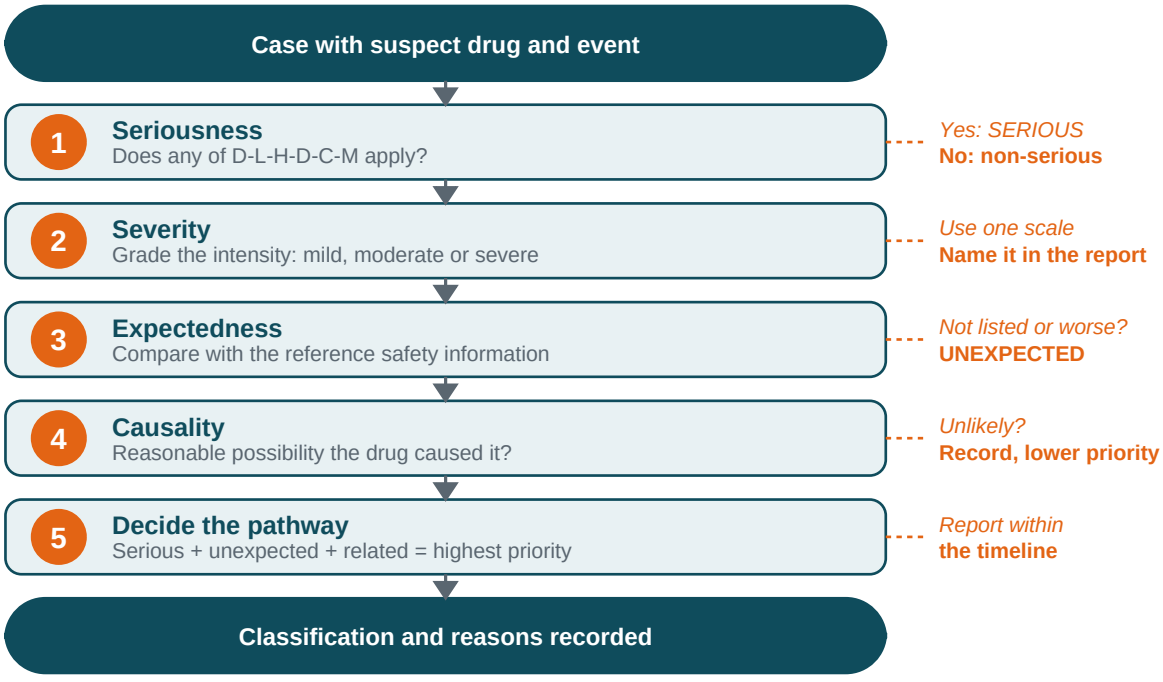
Reporting timelines at a glance

Case type	General expedited timeline
Clinical trial: fatal or life-threatening SUSAR	7 calendar days for the first report, then follow-up within 8 more days.
Clinical trial: other SUSAR	15 calendar days.
Marketed product, EU: serious case (expected or not)	15 calendar days.
Marketed product, US: serious and unexpected	15 calendar days (alert report). Others go in periodic reports.
Marketed product, EU: non-serious case	90 calendar days.

The clock starts on the day the company first receives the minimum information. Rules vary by region and change over time, so always follow local requirements and your SOP.

6. Workflow: Classify Any Case in Four Steps

Hook: **S-S-E-C** = **S**eriousness, **S**everity, **E**xpectedness, **C**ausality. Seriousness comes first because it sets the urgency.



7. Multiple-Case Evaluation

Classify each case yourself first, then compare with the answers. Drugs marked *fictional* are used so that the label extract in the exercise is exact.

Part 1: The cases

Case	Scenario
A	**Lisinopril.** A 58-year-old man develops a persistent dry cough 2 weeks after starting. Lisinopril is changed to losartan and the cough settles. No other treatment, no hospital stay. *Label lists:* cough. *Causality:* Probable.
B	**Amoxicillin-clavulanate.** A 71-year-old woman develops jaundice and itching 3 weeks after a 10-day course. She is admitted for 6 days for investigation (raised liver enzymes) and recovers fully. No other cause found. *Label lists:* cholestatic jaundice and hepatitis. *Causality:* Probable.
C	**Veloxitrin (fictional antibiotic).** A 40-year-old man develops fatigue and dark urine on Day 6, then acute liver failure. He goes to intensive care and is assessed for liver transplant, then slowly recovers. No viral hepatitis, alcohol or other drugs. *Label lists:* nausea, diarrhoea, rash; no liver warning. *Causality:* Probable.
D	**Neurolix (fictional anti-epileptic).** A 33-year-old woman notices mild hair thinning after 8 weeks. No treatment needed and the drug is continued. *Label lists:* dizziness, drowsiness, headache. *Causality:* Possible.
E	**Warfarin.** An 80-year-old man (INR 4.8 two days earlier) falls at home and has an intracranial bleed. He is admitted and dies on Day 3. *Label lists:* bleeding, including fatal haemorrhage. *Causality:* Possible (the fall is an alternative cause).
F	**Valproate.** A woman takes valproate throughout the first trimester of an unplanned pregnancy. Her baby is born with spina bifida. *Label lists:* neural tube defects after exposure in pregnancy. *Causality:* Probable.

Part 2: Classification and clinical impact

Case	Severity	Serious? (criterion)	Expected?	Pathway and impact
A	Mild (Level 2)	No	Expected	Routine; periodic report; watch trends. **Low impact.**
B	Moderate (Level 4)	**Yes**: hospitalisation	Expected	Expedited in the EU (15 days); periodic in the US. **Medium impact.**
C	Severe (Level 5)	**Yes**: life-threatening, hospitalisation	**Unexpected**	**Highest priority.** Expedited alert report (15 days; 7 days if a fatal or life-threatening trial SUSAR), urgent medical review, signal evaluation, likely label update.
D	Mild (Level 1)	No	**Unexpected**	Periodic report; keep for signal detection. **Low impact** unless cases build up.
E	Severe (Level 7)	**Yes**: death, hospitalisation	Expected	Expedited where required; medical review of the case. **High impact** because of the outcome.
F	Severe (Level 6)	**Yes**: congenital anomaly	Expected	Expedited in the EU; follow up the pregnancy outcome. **High impact** (permanent harm to a child).

Impact ranking and teaching points

Rank	Case	Why
1 Highest	C	Serious, unexpected and related: a possible new safety signal that needs urgent action.
2 High	E and F	Serious with the worst outcomes (death, permanent harm), even though both are labelled.
3 Medium	B	Serious and labelled; recovered, but it needed admission.
4 Low	D	Unexpected but mild; its value is as a data point for signal detection.

Rank	Case	Why
5 Lowest	A	Non-serious, expected and mild; routine handling.

- **Expected does not mean unimportant.** Cases B, E and F are all labelled, yet they are serious.
- **Unexpected does not mean serious.** Case D is unexpected but needs only periodic reporting.
- **Severity and seriousness can differ.** Case B is only Moderate (Level 4) but meets the hospitalisation criterion.
- **Causality still counts.** Case E is Possible because the fall is an alternative cause; document that reasoning rather than dropping the case.

8. Writing Up the Classification

Record each dimension, the evidence used, and the resulting pathway in one short block:

> **Seriousness:** Serious (life-threatening, hospitalisation). **Severity:** Severe (Hartwig Level 5). **Expectedness:** Unexpected; hepatic events not listed in the reference safety information in force. **Causality:** Probable (WHO-UMC). **Pathway:** expedited alert report with urgent medical review.

Common mistakes

- Treating **severe** as **serious** (or the reverse).
- Judging expectedness from clinical experience instead of the reference document and its current version.
- Calling a **planned** hospital admission serious.
- Missing the 'more severe or more specific than listed' rule, which makes a labelled event unexpected.
- Discarding non-serious unexpected cases. They feed signal detection.
- Ignoring the **medically important** criterion for events that never reach hospital.
- Not stating which severity scale was used.

The cases are fictional or simplified for training. For real cases follow your organisation's SOPs and current guidance (for example ICH E2A, E2D and E2F, EU GVP Module VI, and local regulations).